

SKILLS

Tools and Languages	Python, Git, LaTeX, C, MySQL, PyTorch, TensorFlow, Pandas, CI/CD Pipelines, AWS, Azure
Research Domains	Machine Learning, Drug Discovery, Generative AI, NLP, LLMs, GNNs, Computational Bio-Chemistry

EDUCATION

BE Computer Science, Birla Institute of Technology and Science, Pilani – Goa Campus	2020-2025
MSc Chemistry, Birla Institute of Technology and Science, Pilani – Goa Campus	2020-2025

TECHNICAL EXPERIENCE

AI Research Scientist <i>MStack AI</i>	Aug 2025 — Present Bangalore, India
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- Designing novel **Chemistry-first AI models** to tackle molecule synthesis and chemical supply chains.
- Engineered multi-GPU distributed training pipelines for novel **chemical foundation models** in reaction understanding.
- Built **chemistry tools and agentic systems** for automated retrosynthesis, condition prediction, and procedure generation.
- Scaled intensive cheminformatics and **Model Context Protocol (MCP) tools** across GPU clusters for high-performance execution.
- Co-authored state-of-the-art research on reaction condition recommendation (**HiRes**) and modular retrosynthesis (**RETROSPECT**, accepted at ICML 2026 AI for Science Workshop).

Research Fellow(DeepRetro Lead) <i>Deep Forest Sciences</i>	Jun 2023 — Aug 2025 Remote
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- Developed tools for **AI-aided retrosynthesis, molecule & materials discovery** and built infrastructure to train **Molecular LLMs**
- Led the design and development of **DeepRetro: a Chemist's AI Co-Pilot for Collaborative Retrosynthesis**. DeepRetro is a novel, **open-source hybrid AI framework** that solves complex retrosynthesis problems.
- Architected a core "**iterative LLM reasoning**" loop integrating the generative power of LLMs with the precision of traditional template-based engines.
- Engineered a **rigorous validation pipeline** to check LLM-generated pathways for chemical validity, stability, and hallucinations, recursively feeding valid precursors back into the search.
- Achieved **state-of-the-art performance**, solving 96.3% of targets on the challenging USPTO-190 multi-step benchmark.
- Demonstrated novel scientific discovery by generating viable, previously **unreported synthetic pathways** for complex natural products like Ohuamine C.
- Migrated models in the open-source drug discovery toolkit **DeepChem** from TensorFlow to **PyTorch**, increasing user base by **92%**

Visiting Researcher Intern <i>Yale University</i>	Jun 2024 — July 2025 New Haven, NY
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- Conducting research on **molecule generation** models using **geometric deep learning** with Prof. Smita Krishnaswamy
- Developing novel algorithms to enhance molecular design efficiency

Research Intern <i>Harvard Medical School</i>	Dec 2023 — Mar 2025 Remote
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- Investigating the neural mechanisms of language processing using **AI models** under **Prof Gabriel Kreiman**
- Applied advanced ML techniques to analyze neural data, contributing to a deeper understanding of language cognition

Student Researcher <i>APPCAIR, Birla Institute of Technology & Science, Pilani</i>	2022 — 2024 Goa
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- Researched the use of **Large Language Models** to generate novel molecules for drug discovery and predict their **synthesizability**
- Devised deep learning methods to predict **protein residue binding** sites
- Evaluated **BERT** and **Transformer**-based architectures for protein residue binding location prediction

PUBLICATIONS

- **Sathyanarayana, S. V.**, Hiremath, S. D., Shah, R., Panda, R., Jana, R., Singh, R., ... & Ramsundar, B. (2026). "DeepRetro: Retrosynthetic Pathway Discovery using Iterative LLM Reasoning". *Nature Scientific Reports*. DOI: 10.1038/s41598-026-38821-z.
- R. Singh, A. A. Barsainyan, R. Irfan, C. J. Amorin, S. He, T. Davis, A. Thiagarajan, S. Sankaran, S. Chithrananda, W. Ahmad, D. Jones, K. McLoughlin, H. Kim, A. Bhutani, **S. V. Sathyanarayana**, V. Viswanathan, J. E. Allen & B. Ramsundar (2026). "ChemBERTa-3: An Open Source Training Framework for Chemical Foundation Models." *Royal Society of Chemistry's Digital Discovery*. DOI: 10.1039/D5DD00348B.
- Pappala, R. S., **Sathyanarayana, S. V.**, Choudhary, R. K., Verma, A., & Warriar, D. (2026). "RETROSPECT: RETROsynthesis via Sequential Prediction, and Chemically Transformed-ranking". *Accepted at the AI for Science Workshop at ICML 2026*.

- **Sathyanarayana, S. V.**, Pappala, R. S., & Warriar, D. (2026). "HiRes: Inspectable Precedent Memory for Reaction Condition Recommendation". *arXiv preprint arXiv:2605.21420*.
- **V. Shreyas**, J. Siguenza, K. Bania & B. Ramsundar "Open Source Molecular Processing Pipeline for Generating Molecules" *Molecular Machine Learning Conference (MoML) 2024, Baylearn 2024, ML4PS workshop at NeurIPS 2024*
- Bisoi, A. V., **Shreyas, V.**, Siguenza, J., & Ramsundar, B. "DeepChem-Variant: A Modular Open Source Framework for Genomic Variant Calling". In *Championing Open-source Development in ML Workshop@ ICML25*.
- Mohanty, D., **Shreyas, V.**, Palai, A., & Ramsundar, B. (2024). Open-source Polymer Generative Pipeline. *4th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE)*, 2025.
- H. Mestha, K. Bania, **V Shreyas**, S. Liu, A. Srinivasan (2025). Multi-Turn Human-LLM Interaction Through the Lens of a Two-Way Intelligibility Protocol. *MTI-LLM Workshop at NeurIPS 2025*.
- **V. Shreyas** and A. Swati, "Predicting ATP binding sites in protein sequences using Deep Learning and Natural Language Processing" *3rd Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE)*, 2024.
- H. Mestha, T. Agrawal, K. Bania, **V. Shreyas** & Y. Bhisikar (2024). "CountCLIP-[Re] Teaching CLIP to Count to Ten". *arXiv preprint arXiv:2406.03586*.
- P. Basa Pati and **V. Shreyas**, "Speech to Equation Conversion using a PoE Tagger," *2022 IEEE 7th International conference for Convergence in Technology (I2CT)*, 2022, pp. 1-4, doi: 10.1109/I2CT54291.2022.9824252.

PROJECTS

Patra — Full-Stack Web Platform (Open Source) 2026
 Built a self-hostable, Git-backed document & static-site publishing platform with authenticated share links and an agent-friendly CLI; a TypeScript monorepo (Hono, React, SQLite) secured with scoped tokens, sandboxed content, SSRF-guarded webhooks and verified by 500+ automated tests.

Metamorph - Generative AI May 2023 — July 2023
 Building an emotionally responsive Humanoid bot with knowledge capture, speech text generation using Generative Large Language Models

ACTIVITIES

Mentor, DeepChem, Google Summer of Code Summer 2024, 2025
 • Mentored Debasish Mohanty and assisted him in adding polymer generative pipelines to the open-source library DeepChem
 • Mentored Bibhusundar Mohapatra and helped him port the Chemception model from TensorFlow to PyTorch

ML Lead, Google Developer Student Clubs, BITS Goa 2022 — 2023
 • Guided a group of 25+ ML enthusiasts, conducted weekly ML training sessions
 • Organized hackathons, mentored in ML model building, and fostered exploration in ML research.

AWARDS AND RECOGNITIONS

Finalist Winner (\$100,000) *Standard Industries Chemical Innovation Challenge* Dec 2024
2nd Runner up | *ACM Goa Chapter's event PitchIt!* Apr 2024

CERTIFICATIONS

Machine Learning *Stanford University, Coursera. May 2022*
Neural Networks and Deep Learning *DeepLearning.ai, Coursera. May 2022*
Natural Language Processing with Deep Learning *Stanford University. May 2022*
Fundamentals of Accelerated Data Science with RAPIDS *Nvidia DLI. Issued July 2021* [Certificate](#)
Python 3 Programming Specialization *University of Michigan, Coursera. December 2020*
Introduction to Artificial Intelligence (AI) *IBM, Coursera. Issued November 2020* [Certificate](#)
What is Data Science? *IBM, Coursera. Issued October 2020* [Certificate](#)